

Final Report: COVID-19 Risk Prediction & Recommendation System

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1. Executive Summary

This project aims to analyze the global impact of the COVID-19 pandemic using data science techniques to predict future risks and optimize resource allocation. The team utilized the Our World in Data (OWID) dataset to build a comprehensive pipeline including data cleaning, Exploratory Data Analysis (EDA), and machine learning modeling.

Key achievements include the development of a regression model to predict smoothed new cases, a classification model to identify "high-risk" days, and a content-based recommendation system to suggest country-specific interventions. Additionally, an interactive Power BI dashboard was created to visualize global trends. The project successfully transitioned from raw data analysis in Week 1 to deploying predictive tools and recommendation logic by Week 3.

2. Introduction

Project Background The COVID-19 pandemic generated vast amounts of data regarding infection rates, mortality, and vaccination progress. Analyzing this data is crucial for governments and healthcare organizations to anticipate surges and allocate medical resources effectively.

Objectives The primary goals of this project were:

- Risk Prediction: Develop machine learning models to forecast infection trends and classify high-risk periods.
- Resource Allocation: Use data-driven insights to recommend proportional resource distribution.
- Similar Country Identification: Implement a recommendation system that groups countries with similar pandemic profiles to share best practices.
- Visualization: Provide an interactive dashboard for real-time monitoring.

Dataset The project utilized the OWID COVID-19 Dataset (`owid-covid-data.csv`), sourced from Kaggle and Our World in Data. This comprehensive dataset includes daily records of cases, deaths, vaccinations, and demographic variables for countries worldwide.

3. Data Preprocessing & Feature Engineering

Data Preprocessing & Feature Engineering

Data Cleaning (Milestone 1) - Missing Values & Duplicates: Duplicates were removed. Missing values were handled using forward-fill methods or median replacement where appropriate. - Standardization: Region labels were standardized, and inconsistent country codes were corrected to ensure accurate geospatial analysis. - Formatting: All date columns were converted to proper datetime objects to facilitate time-series analysis.

Feature Engineering (Milestone 3) - Smoothed Metrics: 7-day moving averages were calculated for cases (`new_cases_smoothed`) and deaths to reduce daily noise and reveal underlying trends. - Per-Capita Metrics: Data was normalized per million population (e.g., `cases_per_million`) to allow for fair comparisons between countries of different sizes. - Growth Rates: Weekly growth rates and `weekly_new_cases` were computed to track the velocity of the spread. - Case Fatality Ratio (CFR): Calculated as `total_deaths / total_cases` to measure the severity of outbreaks. - Categorical Encoding: Features such as continent were encoded for use in machine learning algorithms.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA)

Statistical Analysis - Descriptive Statistics: The team generated summary statistics (head, describe) early in the project to assess the data range and workload. - Hypothesis Testing: Statistical tests were conducted to verify differences in case trends across continents and the relationship between vaccination rates and new cases.

Visualizations - Time-Series: Visualized the progression of cases and deaths over time, comparing different countries. - Correlations: Heatmaps were produced to identify relationships between key variables, such as mobility, vaccinations, and infection rates. - Distributions: Histograms and boxplots helped identify outliers and "unusual trends" in specific regions.

Key Insight: The analysis confirmed that vaccination rates are a strong predictor for reduced case growth, and significant differences exist in outbreak patterns across different continents.

5. Predictive Modeling (Machine Learning)

Predictive Modeling (Machine Learning)

Regression Task Goal: Predict `new_cases_smoothed` to forecast the trajectory of the pandemic. Models Used: - Linear Regression. - Random Forest Regressor. Optimization: Hyperparameter tuning was performed specifically for the Random Forest model to minimize error. Evaluation: Models were evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 scores. Feature importance analysis highlighted the top factors driving the predictions.

Classification Task Goal: Predict a binary target `high_risk_day`, defined based on infection surges. Models Used: - Logistic Regression. - Decision Tree Classifier. Evaluation: Performance was measured using Accuracy, F1-score, and ROC-AUC curves.

6. Recommendation Systems & Resource Allocation

Recommendation Systems & Resource Allocation

Content-Based Recommendation System A custom recommendation engine was built to identify countries with similar pandemic "profiles."

Methodology: The system uses Cosine Similarity and Euclidean distance to compare feature vectors.
Features Used: Case trajectories, vaccination progress, population metrics, and Case Fatality Ratio (CFR). **Output:** For any input target country, the system outputs the top 5 most similar countries. This allows health officials to look at similar nations to see which policy interventions succeeded or failed.

Correlation-Based Resource Allocation The project included a logic layer for "Correlation-Based Resource Allocation," suggesting proportional resource rates based on the predictive model outputs and risk profiles.

7. Power BI Dashboard

Power BI Dashboard

To make the data accessible to non-technical stakeholders, the team developed an interactive Power BI dashboard.

Dashboard Features:

- Global Trends: Line charts displaying global case and death trends over time.
- Regional Comparison: Maps and slicers allowing users to filter by continent or country.
- Vaccination Progress: Visual tracking of vaccination rollout speeds.
- Model Insights: Summaries of the prediction models integrated into the visual layer.

8. Conclusion & Future Work

Conclusion & Future Work

Summary of Results The team successfully executed a full data science lifecycle. From cleaning the OWID dataset to training Random Forest and Decision Tree models, the project met all Milestones 1 through 6.

Key Findings: Vaccination rate proved to be one of the strongest predictors of reduced case growth.

Deliverables: The final submission includes the Python codebase, a trained recommendation system, and a functional Power BI dashboard.

Team Contributions - Kenzhaly Gaziz: Led regression modeling, hyperparameter tuning, feature importance analysis, and the logic for the recommendation system. - Dyuisenov Nurassyl: Managed classification models, cross-validation, hypothesis testing, and validated similarity logic. - Turgunbay Meiirbek: Handled dataset preparation, visualization of model errors (ROC curves), dashboard visuals, and documentation.

Future Steps - Mobility Data: Integrating Google/Apple mobility reports to improve "high_risk_day" prediction accuracy. - Deep Learning: Experimenting with LSTM (Long Short-Term Memory) networks for improved time-series forecasting. - Live Deployment: Hosting the recommendation system as a web API.

9. References

References - Our World in Data (OWID) COVID-19 Dataset. Source: Kaggle / George Saavedra. - Scikit-learn Documentation (used for Linear Regression, Random Forest, Metrics). - Pandas & NumPy Documentation (used for data manipulation). - Power BI Documentation (used for dashboard creation).

Appendix

Appendix A: Implementation Notes & Requirements This appendix lists implementation details, recommended packages, and example commands. Recommended Python packages: - pandas - numpy - scikit-learn - matplotlib - seaborn - reportlab (used to generate PDF) Installation example: `pip install pandas numpy scikit-learn matplotlib seaborn reportlab`

Appendix B: Example Code Snippets (high-level) - Data loading and cleaning: `pandas.read_csv`, `drop_duplicates`, `fillna`, `to_datetime` - Feature engineering: `rolling(window=7).mean()`, `groupby`, `pct_change()` - Modeling: `sklearn.ensemble.RandomForestRegressor`, `sklearn.linear_model.LogisticRegression`